



Deep learning for automated cerebral aneurysm detection on computed tomography images

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Abstract

Purpose Cerebrovascular aneurysms are being observed with rapidly increasing incidence. Therefore, tools are needed for accurate and efficient detection of aneurysms. We used deep learning techniques with CT angiography acquired from multiple medical centers and different machines to develop and evaluate an automatic detection model.

Methods In this study, we have introduced a deep learning model, the faster RCNN model, in order to develop a tool for automatic detection of aneurysms from medical images. The inputs of the model were 2D nearby projection (NP) images from 3D CTA, which were made by the NP method proposed in this study. This method made aneurysms clearly visible on images and improved the model's performance. The study included 311 patients with 352 aneurysms, selected from three hospitals, and 208 and 103 of these patients, respectively, were randomly selected to train and test the models.

Results The sensitivity of the trained model was 91.8%. For aneurysm sizes larger than 3 mm, the sensitivity of successful aneurysm detection was 96.7%. We achieved state-of-the-art sensitivity for > 3 mm aneurysms. The sensitivities also indicated that there was no significant difference among aneurysms at different locations in the body. Computing time for the detection process was less than 25 s per case.

Conclusions We successfully developed a deep learning model that can automatically detect aneurysms. The model performed well for aneurysms of different sizes or in different locations. This finding indicates that the deep learning model has the potential to vastly improve clinician performance by providing automated aneurysm detection.

Keywords Deep learning · Aneurysm detection · CTA · CNN

Introduction

Cerebrovascular aneurysms (CA) have been observed to have high prevalence [1, 2] and are one of the most common causes of subarachnoid hemorrhage (SAH) [3, 4]. Computed tomography angiography (CTA) is a common technique for the detection of aneurysms [5–7]. Usually, it takes time for

radiologists to detect aneurysms from images, particularly those of small size. This can result in the under-detection of aneurysms in clinical practice. Given the risk of life-threatening rupture, it is vital to develop a tool to help radiologists detect aneurysms at an earlier stage.

In previous studies, computer-assisted detection (CAD) systems were proposed to help radiologists find aneurysms [8–10]. Lauric et al. [9] developed an automated intracra-

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nial aneurysm method based on 3D geometry models. They calculated the writhe number for a whole vessel geometry model, and the locations that had an abnormal writhe number were designated as aneurysms. They applied this method to 20 cases and achieved a sensitivity rate of 100% with 5.36 false positives per study using CTA data. Yang et al. [11] developed a CAD system that achieved a sensitivity of 80% (71% for aneurysms less than 5 mm) with three mean false positives per case and 95% (91% for aneurysms less than 5 mm) with nine mean false positives per case. Although these previous studies have demonstrated good performance in terms of aneurysm detection, they were all based on preprocess segmentation technologies, which may limit the possibility of clinical application.

In recent years, deep learning methods have produced remarkable results in the field of object detection. One of the most widely used deep learning methods is the convolutional neural network (CNN), which is often applied to object detection in images. The strength of CNN is its ability to directly utilize raw images as inputs and learn the characteristics of objects during the training process. As a result, it can automatically design characteristics to detect objects, resulting in better performance. Because of this strength, deep learning methods have been applied in many medical fields, including lung cancer prediction [12–14] and diabetic retinopathy detection [15–17]. Therefore, we used deep learning techniques with CT angiography acquired from multiple medical centers and different machines to develop and evaluate a supporting model.

CNN has also been used in aneurysm detection [18, 19]. Stember et al. [19] developed a CNN model for the detection and measurement of cerebral aneurysms on magnetic resonance angiography images. However, the inputs to their model were not whole MR scans. Instead, they used images in which aneurysms were clearly visible to train and test their model. Thus, it would be difficult to use the model for automatic detection of aneurysms based on raw CT or MR scans. To achieve automatic detection, Nakao et al. [20] presented a CNN aneurysm detection system based on maximum intensity projection (MIP) using magnetic resonance angiography (MRA) images. Similarly, Ueda et al. [21] used ResNet-18 CNN to develop an aneurysm detection system with total sensitivities of 91% and 93% for internal and external tests, respectively. However, both studies used fixed images as input to detect aneurysms, and thus, their models would not perform well for aneurysms of different sizes. For example, the sensitivities of aneurysm detection were only 86% and 83% for 5–9.9 mm and ≥ 10 mm aneurysms, respectively, using the model developed by Ueda et al. [21]. Last but not least, most previous studies were based on MR images, with only one deep learning model proposed by Park et al. [22] being based on CTA images. However, their model did not achieve the automatic detection of aneurysms. Since CTA

is a promising candidate as the primary imaging method in the diagnostic workup of intracranial aneurysms [23], it is necessary to develop a model that can detect aneurysms automatically based on CTA images.

For automatic aneurysm detection in clinical applications, the detection model must be able to accept whole scans as input. Furthermore, the detection model should perform well for aneurysms of different sizes, since size varies largely among patients. Therefore, in this study, we have introduced a model that receives raw CTA scans as input and can detect aneurysms with a flexible detection zone to ensure that the model will perform well for aneurysms of various sizes.

Methods

Datasets

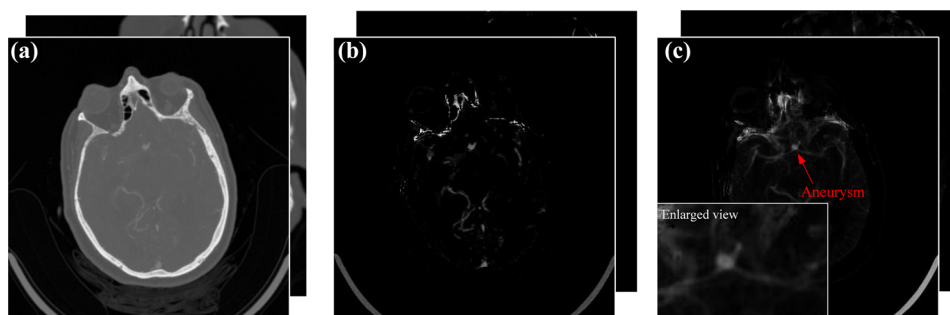
This study involved cerebrovascular CTA images. A total of 311 cases (344 aneurysms) were measured at three medical centers. All image data that are used this study have been approved by Macquarie University human research ethics committees. Each case involved 100–300 CTA slices (DICOM format), depending on slice thickness and the range of each case. All images were found to contain aneurysms and were associated with finalized radiological reports. The radiologists listed as the co-authors annotated independent subsets of aneurysms from their own hospitals according to the radiological reports. Every radiologist in our study had more than 5 years' experience in interpreting CTA. The entire dataset included aneurysms at different locations (see Table 1): 131 aneurysms located at the internal carotid artery (ICA), 95 at the middle cerebral artery (MCA), 35 at the basilar artery (BA), 10 at the vertebral artery (VA), 12 at the posterior cerebral artery (PCA), and 61 at the anterior cerebral artery (ACA). The mean size of all aneurysms was 5.4 ± 3.5 mm (with a range of 1–24 mm).

Among the CTA cases, 283 cases contained a single aneurysm each, while 28 cases contained multiple aneurysms each. The images were obtained with the use of four different CT machines, Toshiba, Siemens, Philips, and GE, with slice thickness of 0.5–5 mm, and were written as DICOM files. All images were 512×512 pixels in resolution. To remove bias in the evaluation of the parameters of the CNN model, all case data were randomly divided into two parts: a training set (comprised of 208 cases) and a test set (103 cases). The training dataset was used to train the model, and the test dataset was used to test the performance of the trained model.

Image pre-treatment

The faster RCNN model was developed to detect objects in 2D images. Cerebrovascular aneurysms, however, are not

Fig. 1 Process of building NP images: **a** original CTA examinations; **b** images with bone removed; **c** final NP images



clearly shown on 2D images (raw CTA slices). In order to introduce the deep learning model into the application of aneurysm detection, it will be necessary to make the aneurysms and arteries appear more clearly on images. So, all CTA images were preprocessed by means of the following steps, which are depicted in Fig. 1. The bright pixels (bone) were removed from the images, and most of the remaining pixels were arteries (Fig. 1b). Next, we used the nearby projection (NP) method to obtain modified images for training and testing the models. Each image was reconstructed by superimposing nearby images with a certain decay rate (Fig. 1c; we refer to the modified images as NP images). The decay rate of the gray value is dependent on the thickness of the CT slices and is calculated as 0.1 times the slice thickness. For example, the decay rate was set as 0.05 per slice for the CT with slice thickness of 0.5 mm.

Model development

With the proposed image pre-treatment above, aneurysms are clearly shown on NP images, which make it possible to utilize a faster RCNN model to detect aneurysms from NP images. A pretrained Resnet-50 [24] model was used to calculate image features in this study. With the calculated image features, the faster RCNN model can propose rectangular boxes on slices that may contain aneurysms. The details of the faster RCNN are available in [25, 26]. The following hyper-parameters were used for training: 20 epochs, learning rate of 5×10^{-5} , and decay rate of 0.8 every 10 epochs. An important characteristic of the faster RCNN model is the ability to detect a target object with changeable rectangular boxes, which enables it to detect aneurysms of various sizes. This feature was an important factor in aneurysm detection in this study, as the size of cerebral aneurysms varied significantly among patients, from 1 to 24 mm in our dataset.

The model was constructed as follows: All NP images with aneurysms in the training set were annotated with the location of each aneurysm. In order to train the CNN model to detect aneurysms, the system was fully supervised using the selected NP images and all annotated aneurysm locations. A set of rectangular boxes that included the aneurysms on

the image and the probability of each being a true positive (aneurysm) was exported by the model (Fig. 2). Horizontal flip was used to augment the training dataset. The training time was about 30 min, a total of 20 epochs with 90 s per epoch.

For deep learning, a phenomenon associated with the training of a model to learn the target characteristic is that the model's performance increases with the size of the training set. We trained three models with 100 cases, 150 cases, and 208 cases, respectively, in the training dataset. These models are referred to as training model 1, training model 2, and training model 3. The total sensitivity was calculated by combining the results of the three trained models.

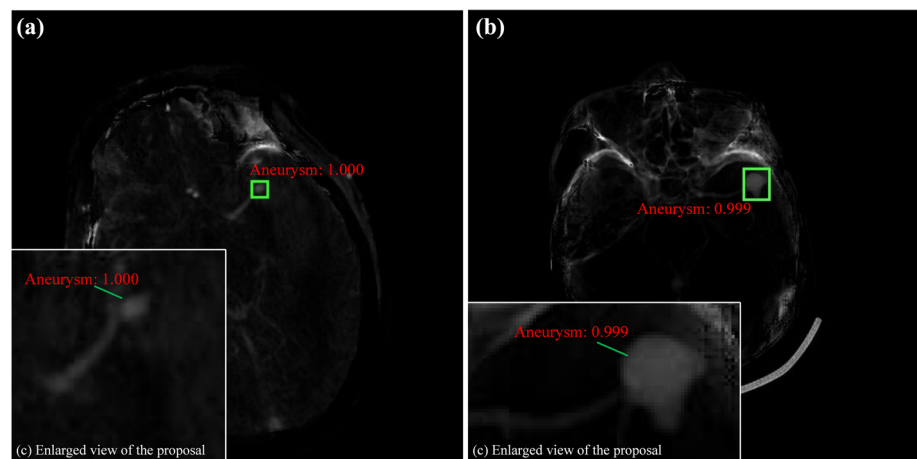
The training process was performed with the use of the Python (2.7.0) programming language and the PyTorch (0.4.0) package on a workstation with one 2080Ti NVIDIA GPU (11 Gigabytes), one 18-core Intel® Core™ i9 7980XE CPU, and 64 GB system RAM. The time required to test each slice was 0.08 s. The total testing time for each case, with 100–300 slices per case, was 8–24 s.

Model evaluation

We used the three trained models to test the sensitivity for all the cases in the test dataset. All image data that are used this study have been approved by local human research ethics committees.

For each case, all slices were input to the trained models, and the probabilities of false positives were calculated. Rectangular boxes were used to mark images with suspected aneurysm locations. True positive detection was determined in the event that the overlapping area between the “proposal” boxes and the ground-truth regions (aneurysm locations) was larger than 30% of the area of the ground-truth regions of the aneurysms. This was the rule for determining whether the rectangular proposal boxes successfully detected the aneurysm. All adjacent slices that detected the same aneurysm were assembled into a single detection instance. Using this rule, we calculated the aneurysm detection sensitivity for the test dataset of 103 cases. We did not use a receiver operating characteristic (ROC) curve to evaluate

Fig. 2 Aneurysm candidates detected by the model in the test dataset (the number represents the probability of the proposal being a true positive (aneurysm) as evaluated by the model)



our model because we were not dealing with a two-classifier problem. All the test samples were positive samples (i.e., they contained aneurysms). Our goal was to determine how many aneurysms could be detected by our model and the number of false positives involved in achieving this sensitivity. Thus, the model performance was evaluated by its sensitivity and the number of false positives. To demonstrate the performance, we created a graph to illustrate the tradeoff between sensitivity and false positives. In addition, we analyzed the sensitivities separately in relation to aneurysm size and location to determine our model performance for different kinds of aneurysms. To evaluate the similarity between the locations of ground-truth aneurysms and the model's proposed locations, we also calculated the Jaccard index (intersection-over-union metric) for each detected aneurysm in the test dataset.

Results

Dataset

The characteristics of the two datasets are shown in Table 1. There were 208 cases with a total of 222 diagnosed cerebral aneurysms in the training dataset. Among all the cases, 194 cases contained a single aneurysm each, while 14 cases contained two or more aneurysms each. The sizes of the aneurysms ranged from 1 to 24 mm (5.4 ± 3.6 mm). The locations of these aneurysms are shown in Table 1: 39.2% at ICA, 26.1% at MCA, 10.8% at BA, 3.6% at VA, 3.2% at PCA, and 17.1% at ACA. The diagnosis of all CTA slices and annotations of aneurysm locations were conducted by experienced radiologists. Table 2 shows the distribution of aneurysms that were randomly selected to train the three models.

The test dataset includes 103 cases with 122 aneurysms (Table 1). Among all the cases, fourteen were found to have two or more aneurysms, while three cases each had an untreated aneurysm and an aneurysm that had been treated

Table 1 Summary of the training and test datasets

	Training dataset	Test dataset
No. of cases	208	103
No. of aneurysms	222	122
<i>Size (mm)</i>		
≤ 3	54 (24.3%)	30 (24.6%)
3–5	76 (34.2%)	44 (36.1%)
5–7	48 (21.6%)	21 (17.2%)
> 7	44 (19.8%)	27 (22.1%)
<i>Location</i>		
Internal carotid artery (ICA)	87 (39.2%)	44 (36.1%)
Middle cerebral artery (MCA)	58 (26.1%)	37 (30.3%)
Basilar artery (BA)	24 (10.8%)	11 (9%)
Vertebral artery (VA)	8 (3.6%)	2 (1.6%)
Posterior cerebral artery (PCA)	7 (3.2%)	5 (4.1%)
Anterior cerebral artery (ACA)	38 (17.1%)	23 (18.9%)
<i>Slice thickness (mm)</i>		
0.5–1	207 (93.2%)	116 (95.1%)
≥ 1.25	15 (6.8%)	6 (4.9%)

(by clipping or coiling). Treated aneurysms were not included in the test process. The aneurysms in the test dataset ranged in size from 1.4 to 18 mm (5.3 ± 3.2 mm) (Table 1). The locations of these aneurysms were as follows: ICA (36.1%), MCA (30.3%), BA (9%), VA (1.6%), PCA (4.1%), and ACA (18.9%). All the CTA slices from the test dataset were obtained from the same machines as the training dataset. As shown in Table 1, there was no significant difference between the training dataset and test dataset in terms of aneurysm size and location.

Overall model performance

We tested three models that had been trained by three datasets: 100, 150, and 208 cases, respectively, in the training

Table 2 Comparison of the three training datasets

	Training model 1 (100 cases)	Training model 2 (150 cases)	Training model 3 (208 cases)
No. of aneurysms	101	157	222
<i>Size (mm)</i>			
≤ 3	25 (24.8%)	39 (24.8%)	54 (24.3%)
3–5	41 (40.6%)	50 (31.8%)	76 (34.2%)
5–7	21 (20.8%)	34 (21.7%)	48 (21.6%)
> 7	14 (13.9%)	34 (21.7%)	44 (19.8%)
<i>Location</i>			
Internal carotid artery (ICA)	38 (37.6%)	53 (33.8%)	87 (39.2%)
Middle cerebral artery (MCA)	24 (23.8%)	44 (28%)	58 (26.1%)
Basilar artery (BA)	13 (12.9%)	18 (11.5%)	24 (10.8%)
Vertebral artery (VA)	5 (5.0%)	7 (4.5%)	8 (3.6%)
Posterior cerebral artery (PCA)	1 (1%)	4 (2.5%)	7 (3.2%)
Anterior cerebral artery (ACA)	20 (19.8%)	31 (19.7%)	38 (17.1%)

Table 3 Correctly detected aneurysms and the sensitivities by three trained models

	Training model 1	Training model 2	Training model 3	Combination (3 models)	<i>p</i> value
Total sensitivity (122 testing aneurysms)	102 (83.6%)	111 (91.0%)	112 (91.8%)	115 (94.3%)	–
<i>Size (mm)</i>					
≤ 3	20 (66.7%)	22 (73.3%)	23 (76.7%)	25 (83.3%)	> 0.05 ^a
3–5	39 (88.6%)	42 (95.5%)	42 (95.5%)	43 (97.7%)	
5–7	20 (95.2%)	21 (100%)	21 (100%)	21 (100%)	
> 7	23 (85.2%)	26 (96.3%)	26 (96.3%)	26 (96.3%)	
<i>Location</i>					
Internal carotid artery (ICA)	38 (86.4%)	39 (88.6%)	39 (88.6%)	41 (93.2%)	> 0.05
Middle cerebral artery (MCA)	28 (75.7%)	34 (91.9%)	34 (91.9%)	34 (91.9%)	
Basilar artery (BA)	9 (81.8%)	10 (90.9%)	11 (100%)	11 (100%)	
Vertebral artery (VA)	1 (50%)	2 (100%)	1 (50%)	2 (100%)	
Posterior cerebral artery (PCA)	5 (100%)	5 (100%)	5 (100%)	5 (100%)	
Anterior cerebral artery (ACA)	21 (91.3%)	21 (91.3%)	22 (95.7%)	22 (95.7%)	

^a ≤ 3 mm aneurysms were not included in Fisher's exact test

dataset. Figure 2 shows two examples of aneurysms detected by the trained models. The detection results from the three trained models are provided in Table 3. For aneurysms in each group by size and location, the sensitivity of aneurysm detection increased as the amount of training data increased, demonstrating that our models were well trained for learning the characteristics of aneurysms in images. Figure 3 shows that the number of false positives tended to remain stable after the sensitivity approached 90%, with mean false positives ranging from 7.9 ± 4.7 to 8.9 ± 5.3 against sensitivities of 90.2% and 91.8% using training model 3. This finding indicates that there were fewer than nine false positives per case on average. After integration of the successful detections from the three models, the total sensitivity was 94.3%, which was higher than the result from training model 3 (91.8%).

Thus, some aneurysms were detected by model 1 or 2 but missed by model 3.

Figure 4 displays the Jaccard index for all the detected aneurysms. The median Jaccard index was 0.55, indicating good similarity of the model's proposals with ground-truth aneurysms. A low index value occurred at some small aneurysms. This happened because the radiologists only annotated the aneurysms on the images, whereas the model annotated both the aneurysms and the arteries.

Impact of aneurysm property

To evaluate the impact of aneurysm size and location on sensitivity, we divided the test dataset into different groups according to size and location. The performance of the model

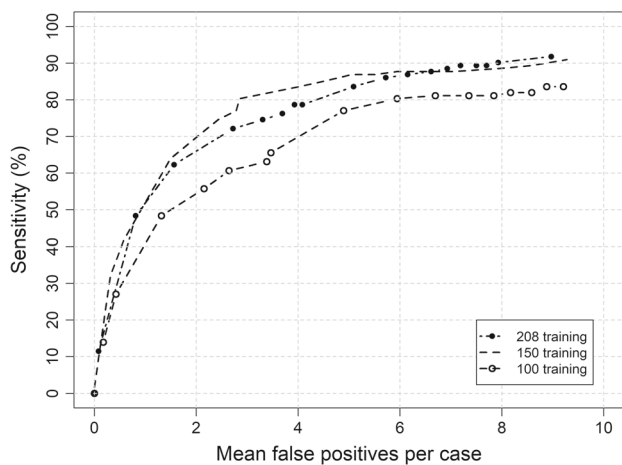


Fig. 3 Mean false positives per case for the test dataset using different trained models

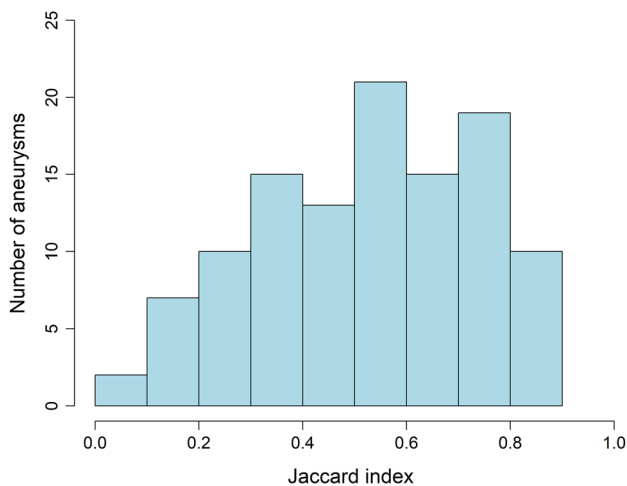


Fig. 4 Distribution of Jaccard index for all detected aneurysms in the test dataset

that successfully detected aneurysms in relation to size and location is shown in Table 3. For training model 3, the sensitivity was 96.7% (89 of 92) for aneurysms larger than 3 mm in size. Fisher's exact test indicated that there was no significant difference in the sensitivity for aneurysms larger than 3 mm. Most of the aneurysms missed by the models were smaller than 3 mm in size. Five small aneurysms (<3 mm) were missed by all three models: three aneurysms located at the MCA and two aneurysms at the ICA.

In regard to the sensitivity at different locations, the sensitivity was higher than 93% for aneurysms at ICA, ACA, PCA and BA, and 91.9% at MCA. Fisher's exact test showed that there was no significant difference in sensitivity for aneurysms at different locations (Table 3).

The detection rate for untreated aneurysms in patients with both an untreated cerebrovascular aneurysm and an aneurysm previously treated by coiling or clipping was 100% (3 of 3).

Thus, the presence of treated aneurysms does not impact the detection of other untreated aneurysms.

Typical aneurysms missed by models and false positives

Seven aneurysms in the test dataset were missed by our models. Among these missed aneurysms, there were similarities. Three aneurysms were smaller than 3 mm and located at the MCA; two aneurysms were smaller than 3 mm and located at the ICA. The other two missed aneurysms were larger than 3 mm and are displayed in Fig. 5: One was a 4-mm ACA aneurysm that looked like a connection vessel between two vessels and another was a 7-mm ICA aneurysm that was located at the lower ICA. Figure 6 shows two typical false positive results, which were both vertical vessels but looked like small aneurysms on NP images.

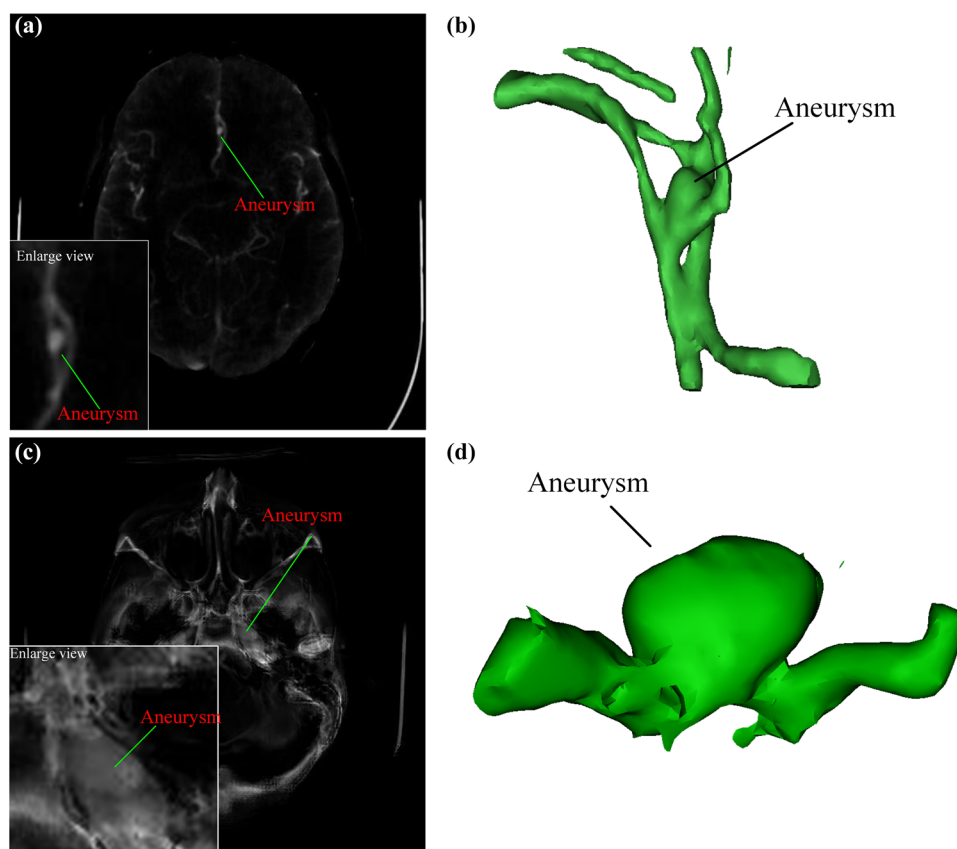
Discussion

In this study, we have proposed a tool for the detection of aneurysms. To our knowledge, this is the first study to achieve automatic aneurysm detection on the basis of CTA images from multiple medical centers using deep learning models. According to the model prediction, all proposals can be automatically marked on 3D view, so that radiologists can easily find the positions on the monitor and check whether their proposals are true or false. An experienced radiologist usually takes 10 to 15 min to complete one aneurysm image diagnosis. By using our model, it will take only a few minutes for radiologist to observe all the proposals, so that they can save almost 10 min per case.

The model successfully detected 91.8% of aneurysms in a test dataset containing 122 aneurysms. Because a flexible decision zone method was used, the model performed well in detecting aneurysms of a wide range of sizes. As shown in Fig. 2, the proposed bounding boxes can automatically fit the aneurysm size. Consequently, our model performed better than those in previous studies in which the size of the decision zone was fixed [20, 21, 27], and we achieved state-of-the-art detection for >3 mm aneurysms. As shown in Table 3, a total sensitivity of 96.7% was attained for the detection of aneurysms with sizes larger than 3 mm, whereas the sensitivity ranged from 83 to 95% for >3 mm aneurysms in [21]. This is an important characteristic in practical application, as larger aneurysms typically carry a greater risk of rupture and are thereby potentially more life-threatening [28–30]. It is therefore important to maximize sensitivity in the detection of aneurysms of different sizes.

The images in both the training and test datasets were obtained with the use of different imaging platforms and slice thicknesses (Table 1). This underscores the appli-

Fig. 5 Examples of aneurysms missed by the model in the test dataset: **a** and **c** results on NP images; **b** and **d** manually built 3D geometry models



capability of our model to datasets obtained from different clinical centers. Moreover, because our algorithm was developed without excluding treated patients, the proposed model is able to detect aneurysms in such patients. This ability is significant because studies have shown that treated patients may be at high risk of rupture of other aneurysms [31].

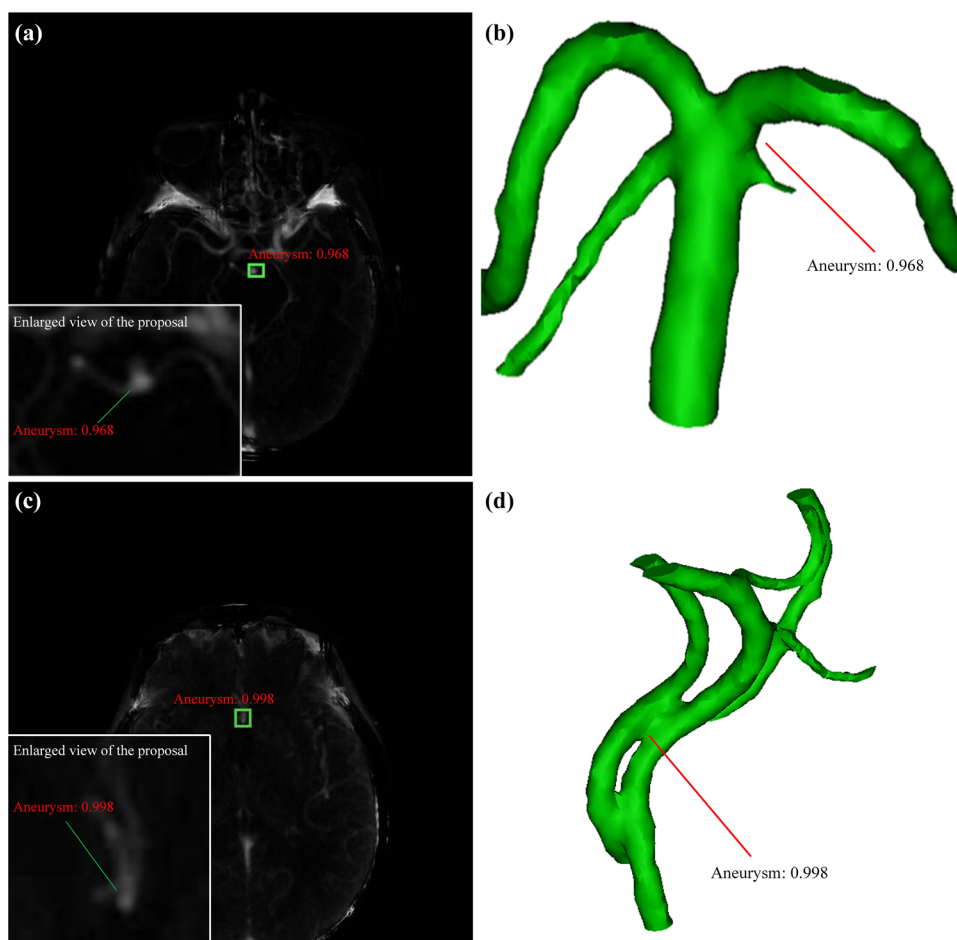
The use of the NP method for CTA images ensures the capability of aneurysm detection with the use of a 2D convolutional network. Our model is thereby much more rapid than models based on a 3D convolutional network. In the present study, the testing process took only about 0.08 s for each slice, and fewer than 25 s were required for completion of the entire detection process for each case.

Our priority in this study was to successfully increase the sensitivity of aneurysm identification and to save time for radiologists to detect aneurysms, so we did not compare the model outcomes with radiologists. We believe that based on current image technologies and support software, experienced radiologists can find almost all aneurysms if they take enough time. The key issue of this study is to save their time and help improve their detection accuracy. Comparing with a previous study, our model directly scans entire patient cases and labels all possible aneurysm locations, whereas the model developed by Ueda et al. [21] used

arterial abnormalities proposed by other software programs as input. Due to this distinction, the number of false positives in our study, per case, may be higher than in previous studies. However, the false positives have been controlled down to an average of nine per case. In addition, we used the training model 3 to test 19 non-aneurysm cases. The number of false positives is 8.6 in average per case. Therefore, at the current stage, radiologists need to further examine these false positive images to determine whether aneurysms are actually present. To further reduce the number of false positives, we will train the model with images that do not contain aneurysms. Alternatively, we could train another model to further distinguish between the true and false positives proposed by the current model. In another aspect, although the mean Jaccard index of 0.55 (the intersection-over-union metric) is not very high, it is sufficient for detection of aneurysms by a radiologist based on the results of our models.

According to our results, the sensitivity was relatively low for aneurysms involving the ICA and for small aneurysms (<3 mm). Some of the aneurysms at the ICA were located near bone. In the current study, a relatively simple method was used to remove the bone pixels from images. However, it was difficult to ensure the removal of all bone pixels and the retention of all pixels of vessels and aneurysms. In the future, we will develop additional training methods to iden-

Fig. 6 Example of false positives suggested by the model in the test dataset: **a** and **c** results on NP images; **b** and **d** manually built 3D geometry models (the number represents the probability of the proposal being a true positive (aneurysm) as evaluated by the model)



tify the bone geometry, thereby increasing the accuracy of bone information.

In regard to issue of patient safety, we believe the sensitivity for aneurysm detection must be 100% to guarantee patient safety. Therefore, the model sensitivity should be improved in further. There was a positive correlation between the size of the training dataset and sensitivity. This indicates that the process of training datasets is steadily improving. In the future, it will be possible to improve the model sensitivity and reduce the occurrence of false positives by including more cases, as well as non-aneurysm cases, in the training dataset. For small aneurysms, the sensitivity may increase with the number of training cases involved. Increasing this number, particularly for small aneurysms in various locations, will be necessary for improving the quality of the models. Furthermore, we will develop our tool by building NP images on both the sagittal and coronal planes to form training images from three directions. The use of images from multiple directions will help to reduce false positives and improve the models' capability in small-size aneurysm detection.

Conclusions

The sensitivity achieved by the model for aneurysm detection was found to be 91.8%. The false positive proposals can be controlled at under nine per case on average. For cases involving aneurysm sizes larger than 3 mm, the sensitivity rate was 96.7%, and the detection process required less than 25 s per case. This study also demonstrated that detection sensitivities were not significantly affected by aneurysm size or location. With ongoing training model improvement, the deep learning model has the potential to improve clinician performance by providing automated aneurysm detection.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was obtained from all individual participants included in the study.

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